

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

CampusNavX: An Intelligent Graph-Based Campus Navigation System for Smart Indoor, Outdoor, Accessible, and Emergency Route Planning Using Advanced Data Structures

Student / Team Information

<i>Team Name</i> <i>Team #</i>	<i>The Pathfinders</i> <i>DSCPP-III-2026-T377</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Walking across an university campus can be hard especially when students and visitors must find certain buildings, classrooms, laboratories, offices or other places. A normal campus map might show where things are. It usually does not help with planning routes moving inside buildings meeting accessibility needs or adapting to changes on campus.

The Campus Navigation System is designed to fix this issue. Campus Navigation System lets users pick a start point and an end point. Then shows them good ways to travel across campus. The Campus Navigation System works inside and outside buildings so people can go between buildings, floors, rooms, laboratories and other places. Campus Navigation System relies on graph diagrams of campus spots and paths to compute routes with BFS, DFS and Dijkstra's algorithm.

The Campus Navigation System also helps when normal maps are not enough. Users can look for paths that skip stairs and use elevators or ramps instead. Campus Navigation System will spot the exits and give the safest or quickest escape routes even when some paths are blocked.

The Campus Navigation System will keep up with these changes. Offer new routes when needed. Campus Navigation System will also keep a record of routes and show clear pictures of the paths and how the algorithms work.

Campus Navigation System shows how data structures and algorithms can be used in life with C++. It uses graphs, queues, priority queues/heaps, linked lists and stacks. That makes it a good choice, for a Data Structures and Algorithms project.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Currently students, staff and visitors mostly rely on campus maps. Ask other people for directions when they need to find a place. These methods are okay for locating buildings and roads but they do not help much when someone needs to find a specific room, laboratory, office or floor, inside a building.

Another problem is accessibility. Regular campus maps usually do not offer route options based on what a person needs. For example someone might want to avoid stairs and instead use an elevator or ramp. Also when a path is blocked or the layout of the campus changes the information may not be updated. In an emergency it can be hard to see the nearest exit or a safe way out.

The proposed system aims to fix these problems by modeling the campus as a graph and using BFS, DFS and Dijkstras algorithm to find the routes. It will support both outdoor navigation include accessible and emergency routes and allow users to save and update route information for later use.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

Project Goals:

1. Develop a C++-based Campus Navigation System for indoor and outdoor navigation across a university campus.
2. Represent the campus using a graph containing buildings, pathways, floors, rooms, laboratories, offices, and facilities.
3. Implement BFS, DFS, and Dijkstra's algorithm for campus traversal and shortest-path finding.
4. Support indoor navigation between rooms and floors while considering stairs and elevators.
5. Provide accessible navigation by avoiding stairs and preferring elevators or ramps.
6. Include emergency navigation to identify exits and provide alternative evacuation routes.
7. Apply Graph, Queue, Priority Queue/Heap, Linked List, Stack, arrays and vectors meaningfully.
8. Provide visualizations of routes, graph traversal, and data structure operations.
9. Support dynamic pathway updates, route history, file handling, dashboards, statistics, reports, validation, and error handling.
10. Use a graphical user interface to make the system interactive and demonstrate the real-world application of DSA.

Project Milestones:

Milestone 1: Design system architecture, campus graph, data structures, and GUI layout.

Milestone 2: Implement campus locations, pathways, and core data structures.

Milestone 3: Implement BFS, DFS, Dijkstra, and route reconstruction.

Milestone 4: Develop indoor and outdoor navigation with floors, rooms, stairs, and elevators.

Milestone 5: Add accessible and emergency navigation with alternative routes.

Milestone 6: Integrate visualization, route history, dynamic updates, file handling, dashboards, and reports.

Final Milestone: Complete testing, complexity analysis, algorithm comparison, error handling, GUI refinement, and final demonstration.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

The new Campus Navigation System is going to be built using C++. Will have a graphical user interface. The purpose is to make it easy and fun to move around the campus. The campus will be shown as a graph, with buildings, rooms, floors, labs and offices as points and the paths between them as lines. These lines will have details like how they are, how long they take to walk and if they are open.

The system will have parts. One part will handle the locations, like all the buildings, floors, rooms, labs and other places. Another part will help people find their way outside using methods, like BFS, DFS and Dijkstras algorithm. These paths will appear on the screen in a way. There will also be a part that helps people avoid stairs and use elevators or ramps instead. Another part will help in case of emergencies by pointing out exits and finding ways to get out if the usual paths are blocked.

The system will use graphs, queues, priority queues, linked lists, stacks, arrays and vectors. It will show step by step how these things work. The system will also be able to change when the campus changes keep a record of routes let people go back handle files find locations, filter results, show dashboards give stats make reports check inputs and deal with errors. C++ will be the tool and the exact way to make the graphical user interface will be decided later based on what is needed.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice.

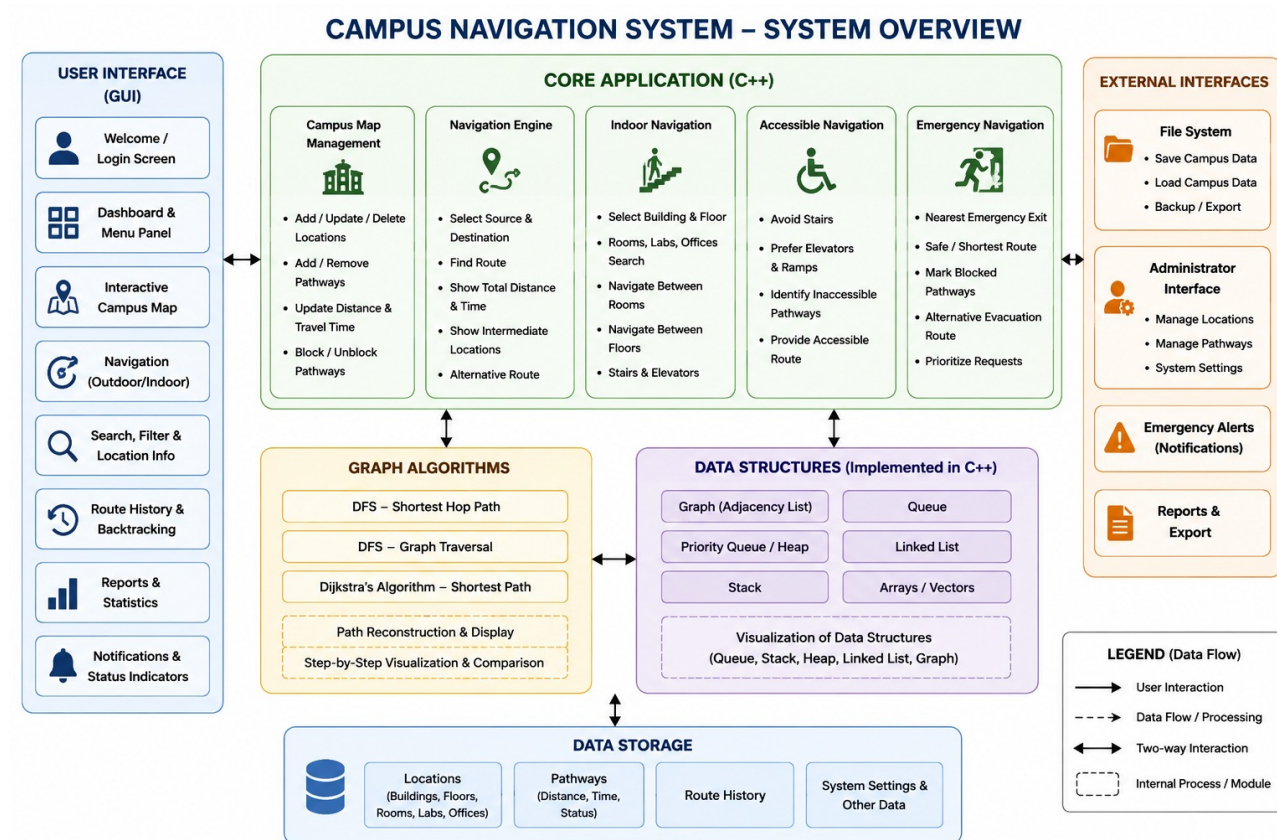


Figure 1: System Overview of the Campus Navigation System

Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

Project Deliverables

1. Campus. Navigation:

I plan to create a map of the campus using graphs to show buildings, rooms, floors, paths and key facilities. Campus Map and Navigation will help you find your way both indoors and outdoors. Use of the algorithm for finding the routes.

2. Updates, History and Performance Analysis: This feature will allow you to update the campus paths and will maintain a log of all the paths along with the navigation data. It will also provide you with statistics. By using the Updates, History and Performance Analysis you will be able to see the comparison between different algorithms and data structures. Also accessible routes will show and will also include GUI.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

Pathways have defined distances, estimated times to travel and a status that indicates whether they are open or closed. The system prompts the user to supply a starting location and a destination. The system stores information concerning the accessibility of places such as stairs, elevators, ramps and paths that are not navigable. For emergency navigation, the system contains information on emergency exits and blocked pathways. The system updates campus data whenever a location or a pathway is modified.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

- 1. cppreference — Reference for C++ language features, standard library, and programming concepts.
Link : <https://en.cppreference.com>*
- 2. GeeksforGeeks – Data Structures and Algorithms — Reference for graphs, BFS, DFS, Dijkstra's algorithm, stacks, queues, and linked lists.
Link : <https://www.geeksforgeeks.org>*
- 3. Programiz – Data Structures and Algorithms — Reference for fundamental data structures and graph algorithms.
Link : <https://www.programiz.com>*
- 4. OpenStreetMap — Reference for understanding map-based representations of locations, pathways, and geographic navigation.
Link : <https://www.openstreetmap.org>*